Introduction to Machine Learning - Sample Notes

Machine learning (ML) is a branch of artificial intelligence in which computers learn patterns from data instead of being given step-by-step rules. Artificial intelligence is the broad goal of making machines behave intelligently. Machine learning is one way of achieving that goal. Deep learning is a smaller part of machine learning that uses neural networks with many layers.

There are three main types of machine learning. In supervised learning, the model learns from labelled examples, where every input has a known correct output. Linear regression and decision trees are common supervised algorithms. Regression predicts a number, such as a house price, while classification predicts a category, such as spam or not spam.

In unsupervised learning, the data has no labels and the model looks for hidden structure on its own. K-means clustering is a popular unsupervised algorithm that groups similar items together. In reinforcement learning, an agent learns by trying actions in an environment and receiving rewards or penalties.

Training, Testing and Overfitting

To check whether a model really learned something useful, the data is usually split into a training set and a test set. A common split is 80 percent for training and 20 percent for testing. The model learns from the training set, and the test set is kept hidden until the end to measure how well the model works on new data.

Overfitting happens when a model memorizes the training data, including its noise, and then performs badly on new data. Underfitting is the opposite problem: the model is too simple to capture the pattern, so it performs badly even on the training data. A good model balances the two.

Cross-validation is a technique that reduces the risk of a misleading result. The data is divided into several parts called folds. The model is trained and tested several times, using a different fold as the test set each time, and the scores are averaged. Overfitting can also be reduced by collecting more data or by using a simpler model.

Evaluation Metrics and Data Preprocessing

Accuracy is the fraction of predictions that are correct. Accuracy can be misleading when one class is much bigger than the other. For example, if only 1 percent of emails are spam, a model that always says "not spam" is 99 percent accurate but useless.

Precision is the fraction of predicted positives that are truly positive. Recall is the fraction of real positives that the model manages to find. The F1 score is the harmonic mean of precision and recall, so it is high only when both are high. A confusion matrix is a table that counts true positives, false positives, true negatives and false negatives.

Real-world data is messy, so preprocessing is needed before training. Missing values can be filled using the mean, the median or the most common value. Feature scaling puts numbers on a similar range so that large values do not dominate. Categorical values such as city names are converted into numbers using methods like one-hot encoding.